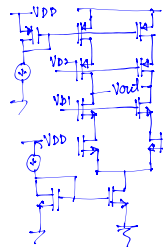
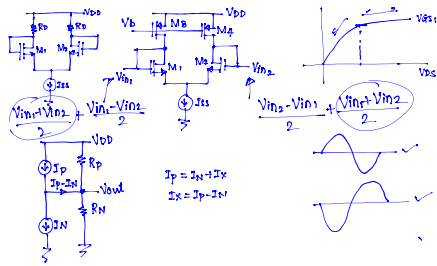
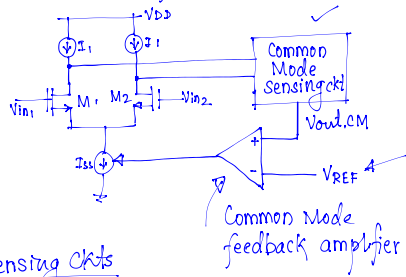


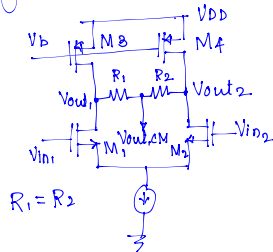
Common Mode Feedback:-



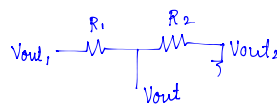
General structure of CMFB



Sensing Ckts



Gain without R_1 and R_2
 $= g_{m1}(r_{o1} || r_{o3})$
 Gain with R_1 and R_2
 $= g_{m1}(r_{o1} || r_{o3} || R_1)$

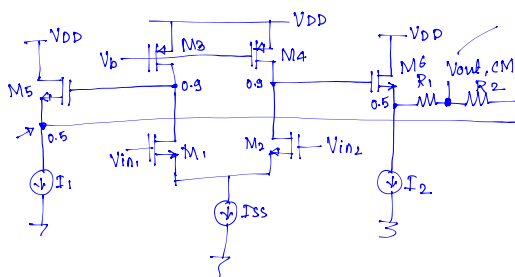


$$V_{out1} = V_{out1} \frac{R_2}{R_1 + R_2}$$

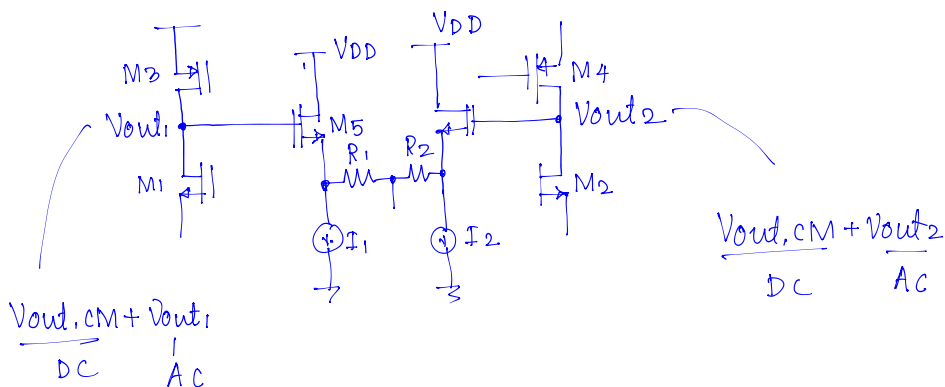
$$V_{out2} = V_{out2} \frac{R_1}{R_1 + R_2}$$

$$V_{out,CM} = V_{out1} \frac{R_2}{R_1 + R_2} + V_{out2} \frac{R_1}{R_1 + R_2}$$

$$= \frac{V_{out1} + V_{out2}}{2}$$



$$\frac{V_{out1} + V_{out2}}{2} - V_{GSS,6}$$



$$\frac{V_{out,CM} + V_{out1}}{DC}$$

$$\frac{V_{out,CM} + V_{out2}}{AC}$$